Regression Discontinuity

ECON726

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Housekeeping

Today: regression discontinuity design

- Largely inspired by Causal inference mixtape
- Some examples taken from MHE companion website

Next week: Instrumental variables

- Chapter 4 of MHE, if your paper uses instrumental variables
- Otherwise, read: Causal inference mixtape

Two weeks: Guest lecture!

- Zachary Feder, the Open Data Program Manager (and my colleague!), will talk about NYC Open Data

Empirical project

Paper selection: Due **September 22** [5% of final grade]

- Done!

Project plan: Due **October 14** [15% of final grade]

- Done!

Replication results: **November 7**: [10% of final grade]

- You should submit some of your main replication results, uploaded to GitHub.

Presentation: Due **December 2** & **December 9** [10% of final grade]

- Prepare a 5-8 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 16** [35% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Outline

Background

Sharp Regression Discontinuity Design

Sharp Regression Discontinuity Design Estimation

Fuzzy Regression Discontinuity Design

Challenges

Background

Sharp Regression Discontinuity Design

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Fuzzy Regression Discontinuity Design

Challenges

History of RDD

History of regression discontinuity design (RDD) reviewed in Cook (2008)

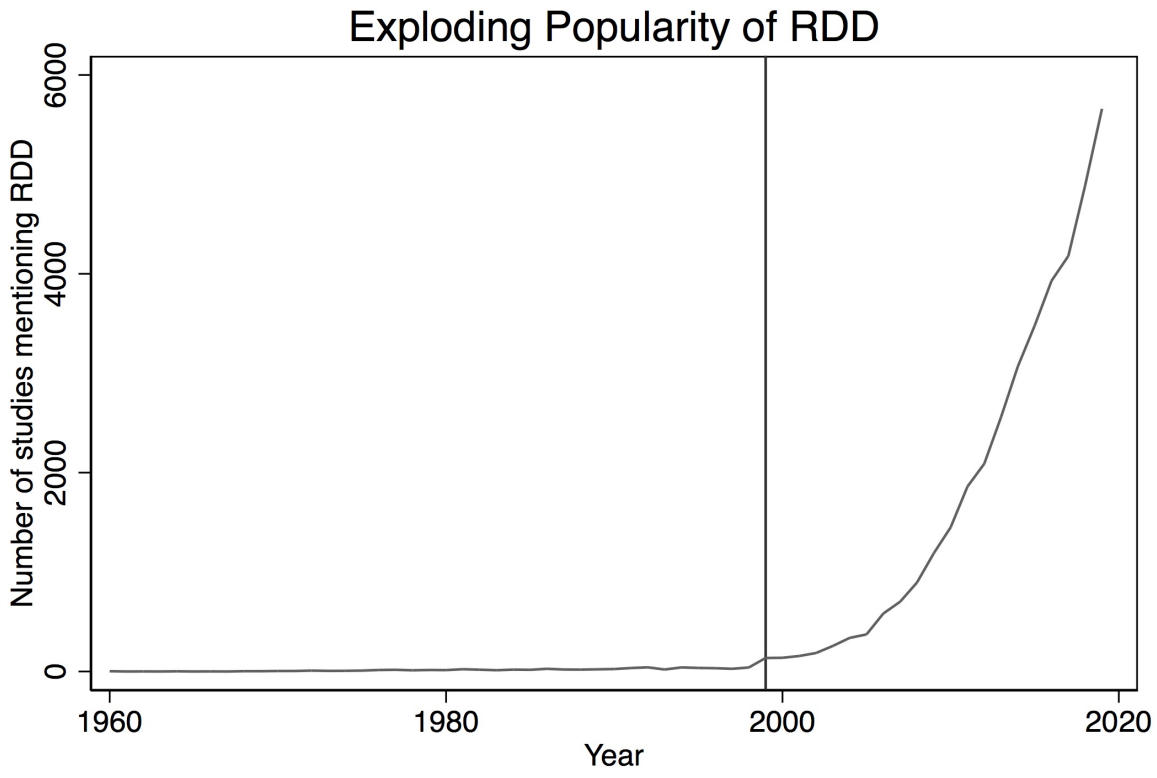
First developed by two psychologists in Thistlewaite and Campbell (1960)

- Fell out of use in psychology
- Never caught on with Statisticians
- Introduced in Economics in 1972, but had a rather poor reception until about 1995

In 1999, several papers published that used RDD:

- Angrist and Lavy (1999): Look at cutoffs in classroom size in Israel to measure the impact of class size on academic outcomes
- Black (1999): school zone geographic discontinuities to estimate parents' willingness-to-pay for better schools

RDD over time



Why the lag in adoption?

Cook (2008): RDD was "waiting for life to arrive"

- In the 1980s, economists often used models and theory to resolve issues of selection bias

  Some [economists] were taken with instrumental variable (IV) approaches that depend on discovering instruments correlated with selection but not with errors in the outcome... It was one thing to have a highly abstract, elegant and general theory about unbiased causal inference; and quite another thing for this method to generate specific applications whose causal conclusions fellow scholars saw as beyond dispute insofar as all the central assumptions were manifestly met.

- Credibility revolution in Economics
  - Acceptance of the potential outcomes framework
  - Increased emphasis on causality

- Availability of large administrative datasets

Example of RDD

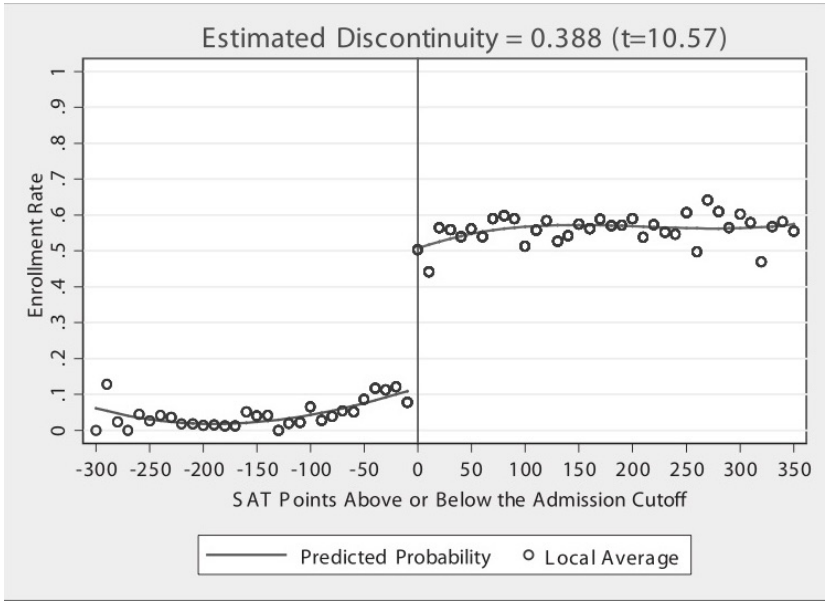
Hoekstra (2009): Wanted to measure the effect of attending a flagship state university on the earnings of 28 to 33 year olds

- Measures the heterogeneous returns to college attendance
- State flagships are generally more selective than other state schools
- Example: Florida
  - University of Florida (UF): state flagship
  - Other public universities in Florida: University of Central Florida (UCF); University of South Florida (USF); Florida International University (FIU); Florida State University (FSU)...

Example: what is the benefit of attending the state's flagship program?

- Selection bias: students who selected into the flagship school may be higher "ability;" this results in them having higher wages than those who did not attend the flagship school

Attending the state flagship university as a function of re-centered standardized test scores



1. Horizontal access is "re-centered" SAT score. The vertical line marks the cut-off (minimum SAT score for admissions)
   - Re-centered SAT score is the **running variable**
2. Dots represent conditional mean enrollments per re-centered SAT score for many students
3. There are two separate curvy lines. These lines are fit separately to the left and right of the cutoff
4. There is a jump at zero in the running variable;
   - The probability of enrolling at the flagship state university jumps discontinuously when the student has the minimum SAT score required by the school

Interpretation of discontinuity

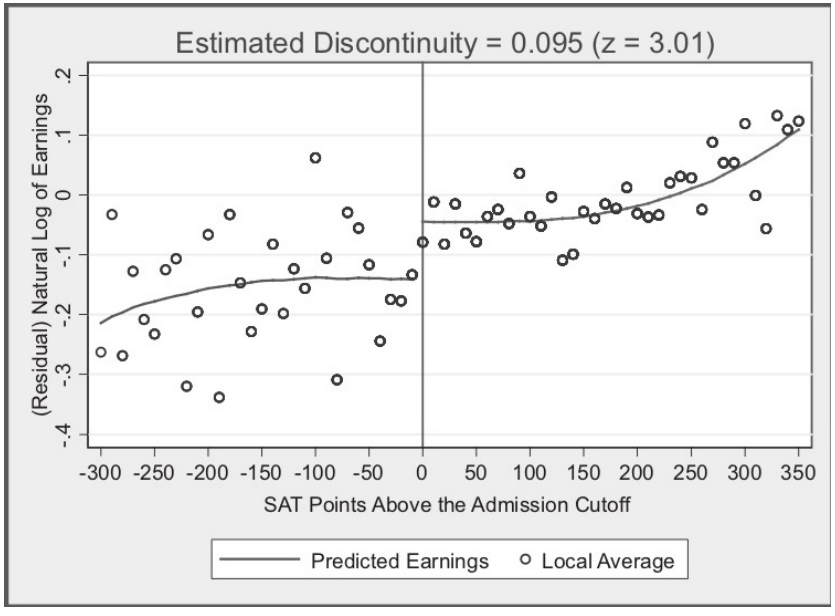
Suppose the admission cut-off is 1250,

$\Longrightarrow$ student with a 1250 has a much higher probability of attending than a student with a 1240

Individual student with a 1250 and individual student with a 1240 might be similar.

- Each dot in this figure represents hundreds of students
- Motivating assumption for RDD: how different is the average student with a 1240, and the average student with a 1250? Is there a meaningful jump in ability at that level?

Estimation



Impact of flagship on earnings:

- Estimate individual characteristics on earnings

$$\text{log Earnings} = \psi_{\text{Year}} + \omega_{\text{Experience}} + \theta_{\text{Cohort}} + \epsilon \quad (1)$$

  Save residuals from (1)

- Average residual earnings along bins for running variable

Interpretation of results

"Estimated Discontinuity = 0.095 (z = 3.01)"

- Those just above the cutoff earn 9.5% higher wages in the long term than do those just below.

Intuitive understanding

- Students with an SAT score just above the cutoff see a jump their probability of attending the state flagship
- Ten+ years later, these students just above the SAT cutoff see a jump in logged earnings of 9.5%

$\Longrightarrow$ Those students who barely made it into the state flagship made around 10% more in long term earnings

Discontinuities

RDD is about finding "jumps" in the probability of treatment along some running variable X

Assignment rule does not need to be arbitrary; it just needs to be known, precise, and free of manipulation. Examples:

- Hansen (2015): To study the effect of harsher punishments and sanctions on driving under the influence, the authors use 0.08 BAC threshold, which increases the probability of being arrested for DWI
- Card, Dobkin, and Maestas (2008): To study the impact of health insurance on health care utilization, the authors use the onset of Medicare eligibility at 65
- Jacob and Lefgren (2004): To study the effectiveness of remedial education programs like summer school, the authors use a grade threshold that increases the probability of summer school attendance

In all cases: need lots of data around the discontinuities. RDD studies often use large, administrative datasets

Background

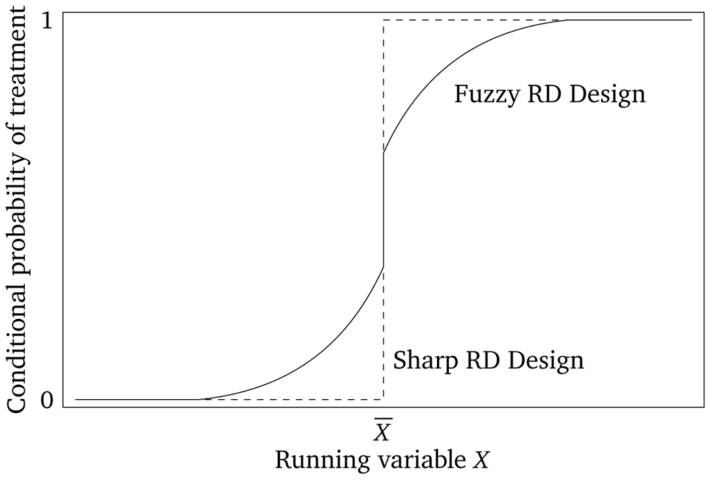
Sharp Regression Discontinuity Design

Sharp Regression Discontinuity Design Estimation

Fuzzy Regression Discontinuity Design

Challenges

Two types of RDD designs



Sharp RDD:

- Probability of treatment goes from 0 to 1 at cutoff
- Treatment is a **deterministic function** of running variable X
- Example: Medicare enrollment happens at 65

Fuzzy RDD:

- Discontinuous "jump" in probability of treatment at cutoff
- Cutoff is used as an instrumental variable for treatment (more on this later)

In both sharp and fuzzy RDD: there is some running variable X, where, upon reaching some cutoff c_0, the likelihood of treatment flips

Sharp RDD

Treatment D_i is a deterministic function of the running variable X_i:

$$D_i = \begin{cases} 1 & \text{if } X_i \geq c_0 \\ 0 & \text{if } X_i < c_0 \end{cases}$$

where c_0 is a **known** threshold or cutoff

Implications of sharp RDD:

- If you know X_i, you know D_i
- There is no **overlap** along the running variable

Potential outcomes in Sharp RDD

Assuming constant treatment effects in sharp RDD, can write control potential outcome as:

$$Y_i(0) = \alpha + \beta X_i$$
$$Y_i(1) = Y_i(0) + \delta,$$

where δ is the treatment effect.

Can write observed Y_i in terms of potential outcomes:

$$Y_i = Y_i(0) + (Y_i(1) - Y_i(0))D_i$$
$$\implies Y_i = \alpha + \beta X_i + \delta D_i + \epsilon_i$$

Treatment effect parameter δ is the discontinuity in the conditional expectation function:

$$\delta = \lim_{x \downarrow c_0} \mathbf{E}[Y_i(1)|X_i = x] - \lim_{x \uparrow c_0} \mathbf{E}[Y_i(0)|X_i = x]$$

Local average treatment effect (LATE)

Causal effect in sharp RDD at the cutoff is the **local averate treatment effect (LATE)**:

- Average causal effect as running variable approaches the cutoff in the limit
- Technically only identifying causal effects for those at the limit
- We have only estimated an average treatment effect that is **local** to the range around the cutoff

$$\delta_{SRD} = \mathbf{E}[Y_i(1) - Y_i(0)|X_i = c_0]$$

In sharp RDD, **extrapolation** plays an important role

- There is no overlap in treatment before and after the cut-off

Notebook: generate simulated data

Continuity assumption

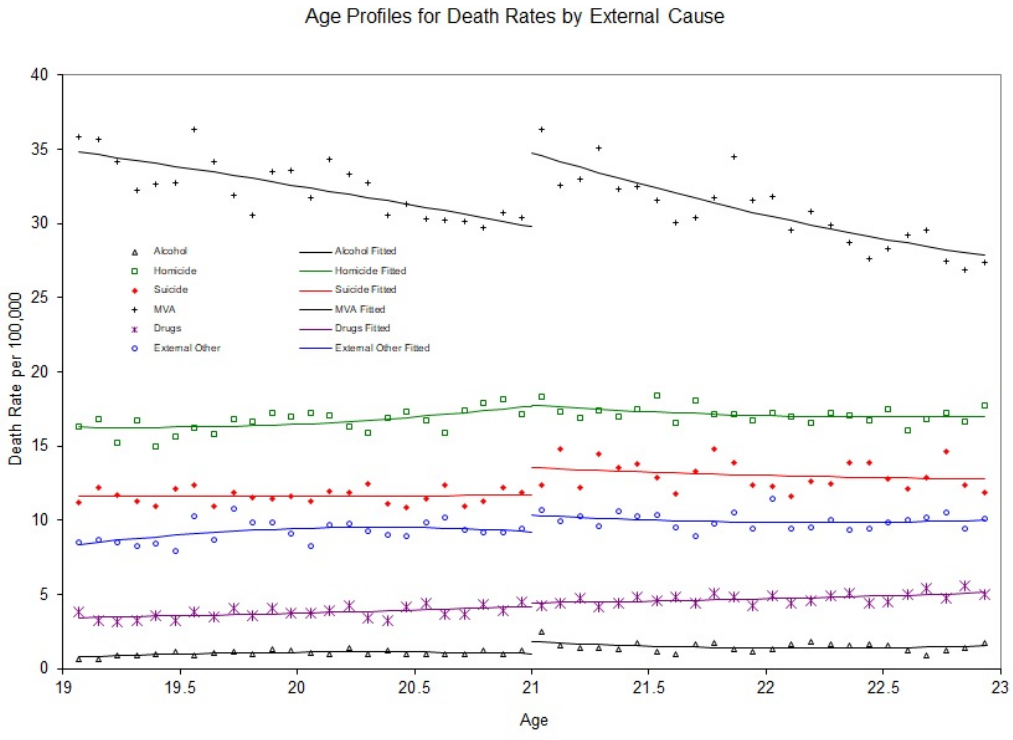
Key assumption in Sharp RDD: continuitiy of potential outcomes

- $\mathbf{E}[Y_0|X = c_0]$ and $\mathbf{E}[Y_1|X = c_0]$ are **continuous functions** of X, even across the threshold c_0
- Absent treatment, potential outcomes wouldn't have jumped
- **Fundamentally untestable assumption**

Example: Carpenter and Dobkin (2009)

- Effect of alcohol consumption on mortality (specifically car crashes)
- If there's a discontinuous jump in drinking at 21, and that causes more people to drive drunk, that may result in more deaths from motor vehical accidents (MVAs)
- Continuity assumption: nothing else could happen that causes mortality of 21 year olds to jump

Continuity assumption: example



- There is a discontinuous jump in motor vehicle deaths at 21
- Most likely explanation is that age 21 causes people to drink more, and sometimes even while they are driving
- This is only a causal effect if motor vehicle accidents don't jump at age 21 for other reason

Example notebook

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Challenges

Centering running variable

Common for authors to "re-center" running variable X_i by subtracting the cut-off, c_0.

$$
\begin{aligned}
& Y_i = \beta_0 + \beta_1 X_i + \delta D_i + \epsilon_i \\
\implies \; & Y_i = \beta_0 + \beta_1 (X_i - c_0) + \delta D_i + \epsilon_i \\
& = (\beta_0 - \beta_1 \cdot c_0) + \beta_1 X_i + \delta D_i + \epsilon_i \\
& = \alpha + \beta_1 X_i + \delta D_i + \epsilon_i
\end{aligned}
$$

Variables can maintain the same interpretation if we re-center the running variable X_i and change the intercept

Linear Sharp RDD

Goal: estimate treatment effect at the cut-off, c_0

- Many sharp RDDs are nonlinear, but it's helpful to start thinking about estimation in a linear model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 \mathbf{1}\{X_i > c_0\} + \beta_3 \mathbf{1}\{X_i > c_0\} X_i$$

- Estimating this equation is the same as estimating a linear model above and below the threshold
  - β_0 is intercept before cut-off
  - $\beta_0 + \beta_2$ is intercept after cut-off

Why do we re-center the running variable X_i?

- If you re-center the model, β_0 is the predicted value at the threshold, for the values below c_0

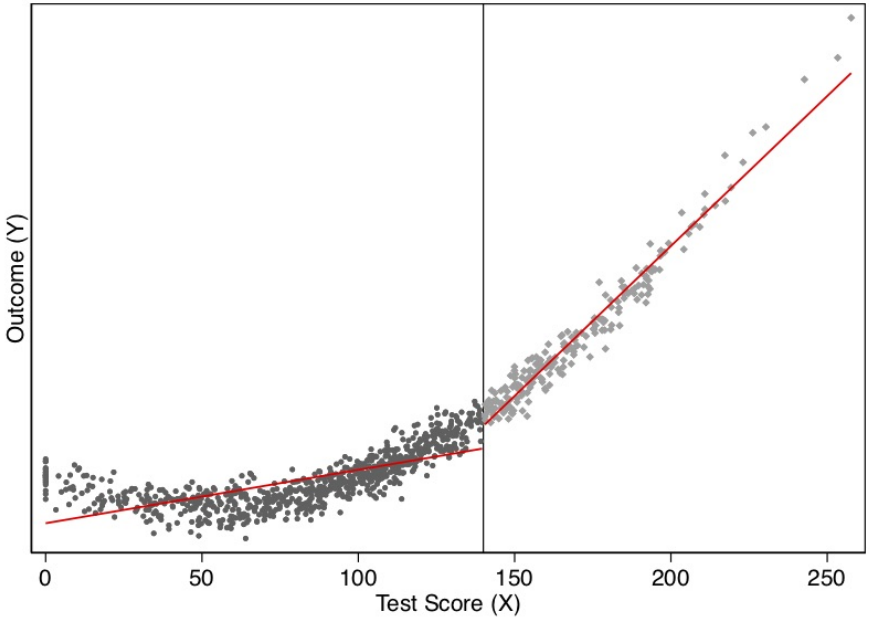
$$\beta_0 = \lim_{x \uparrow c} \mathbf{E}[Y|X_i = x]$$

- Then $\beta_0 + \beta_2$ is the predicted value at the threshold, for values above c_0

$$\beta_0 + \beta_2 = \lim_{x \downarrow c} \mathbf{E}[Y|X_i = x]$$

$\Longrightarrow$ ATE equals β_2

Nonlinearities



- True data generating process is nonlinear
- Using a linear model may lead to a spurious example
- On the left: fitting a model using a least squares regression controlling for the running variable estimates a causal effect though there isn't one

Nonlinearities

Suppose the nonlinear relationship is:

$$\mathbf{E}[Y_i(0)|X_i] = f(X_i),$$

where $f(X_i)$ is a reasonably smooth function of X_i. Then fit the regression model:

$$Y_i = f(X_i) + \delta D_i + \epsilon_i$$

Question: since $f(X_i)$ is unobservable for $X_i > c_0$, how do you model the nonlinearity?

1. Estimate $f(X_i)$ using a p-th order polynomial
2. Nonparametric kernel estimation

Nonlinear estimation: polynomial

Generate $f(X_i)$ by allowing X_i to differ on both sides of the cutoff. Letting $\tilde{X}_i$ be the centered running variable:

$$\mathbf{E}[Y_i(0)|X_i] = \alpha + \beta_{01}\tilde{X}_i + \beta_{02}\tilde{X}_i^2 + \cdots + \beta_{0p}\tilde{X}_i^p$$
$$\mathbf{E}[Y_i(1)|X_i] = \alpha + \delta + \beta_{11}\tilde{X}_i + \beta_{12}\tilde{X}_i^2 + \cdots + \beta_{1p}\tilde{X}_i^p$$

Translating this into observed Y_i:

$$\mathbf{E}[Y_i|X_i] = \mathbf{E}[Y_i(0)|X_i] + \Big(\mathbf{E}[Y_i(1)|X_i] - \mathbf{E}[Y_i(0)|X_i]\Big)D_i$$

Regression model given by:

$$Y_i = \alpha + \beta_{01}\tilde{X}_i + \beta_{02}\tilde{X}_i^2 + \cdots + \beta_{0p}\tilde{X}_i^p + \delta D_i + \beta_1^* D_i\tilde{X}_i + \beta_2^* D_i\tilde{X}_i^2 + \cdots + \beta_p^* D_i\tilde{X}_i^p + \epsilon_i$$

where $\beta_p^* = \beta_{1p} - \beta_{0p}$.

- At c_0, treatment effect given by δ
- At $c > c_0$, treatment effect given by $\delta + \beta_1^* c + \beta_2^* c^2 + \cdots + \beta_p^* c^p$

Nonlinear estimation: polynomial

Validity of RDD estimates using polynomials depends on whether polynomial model provides an accurdate description of $\mathbf{E}[Y_i(0)|X_i]$

- If polynomial model is not appropriate, then what looks like a jump might actually just be unaccounted for nonlinearity

Gelman and Imbens (2019): "Why High-Order Polynomials Should Not Be Used in Regression Discontinuity Designs"

- Controlling for higher-order polynomials (3+) can lead to nonsensical results
- Three major problems:
  - Noisy estimates
  - Sensitivity to degree of the polynomial
  - Poor coverage of confidence intervals
- Authors recommend local linear or quadratic polynomials or other smooth functions

That said, it's conceptually important, and used in older papers!

Nonlinear estimation: kernel estimation

Kernel regression provides an alternative to using higher-order polynomials

Regression Discontinuity relies heavily on the extrapolation

- Looking at the values at the beginning and end of 2 regression lines
- If regression is focusing on fitting other data points, rather than correctly fitting at the threshold, the treatment effect might be misspecified

Possible solution: give higher weights to points close to the cutoff c_0

Hahn, Todd, and Van der Klaauw (2001):

- To avoid poor boundary performance, use weighted regression restricted to a window ("local")
- In local kernel regression: "kernel" provides weights close to the cut-off; can then use weighted least squares to estimate effects

Kernel weighting function

Re-weight sample using the **triangle kernel**:

$$K(X, c_0, h) = \mathbf{1}\{|X - c_0| \leq h\} \cdot \left(1 - \frac{X - c_0}{h}\right)$$

- $|X - c_0| \leq h$: The absolute difference between X and c_0 is less than or equal to some **bandwidth** h
- $\mathbf{1}\{|X - c_0| \leq h\}$: Indicator function that equals 1 if X is sufficiently close to c_0, according to the bandwidth h; 0 otherwise
- $\left(1 - \frac{X - c_0}{h}\right)$: re-weighting function
  - Ignore cases where $X - c_0 > h$, because of the indicator function
  - We know $(X - c_0)/h$ is less than or equal to 1
  - The further X is from c, the larger this fraction is
  - Because this fraction is subtracted from 1, the weights get smaller the farter X is from the cut-off

Kernel regression

Once you've established kernel weighting function, you can use **weighted least squares** to estimate regression

$$(a, b) = \arg\min_{a,b} \sum_{i=1}^{n} (Y_i - a - b(X_i - c_0))^2 K(X, c_0, h)$$

Note:

- This method is sensitive to the choice of bandwidth h. There is a lot of work out there about determining the proper bandwidth!

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Challenges

Fuzzy RDD

Sharp RDD: treatment determined when $X_i \geq c_0$

Sometimes discontinuity is not deterministic

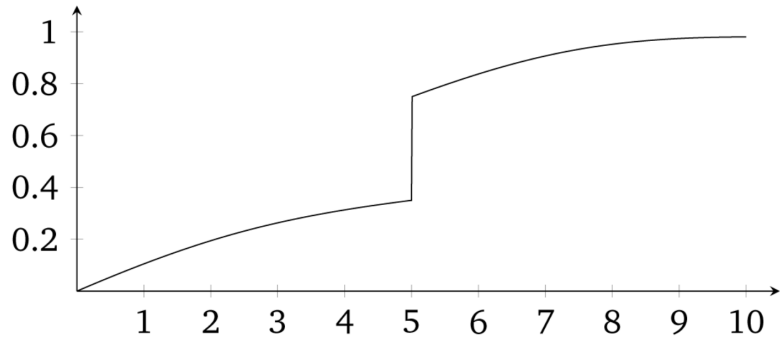
- Fuzzy RDD: increase in the probability of treatment assignment
- May happen if incentives change at some threshold, which may move some, but not all, observations to treated

In terms of limits:

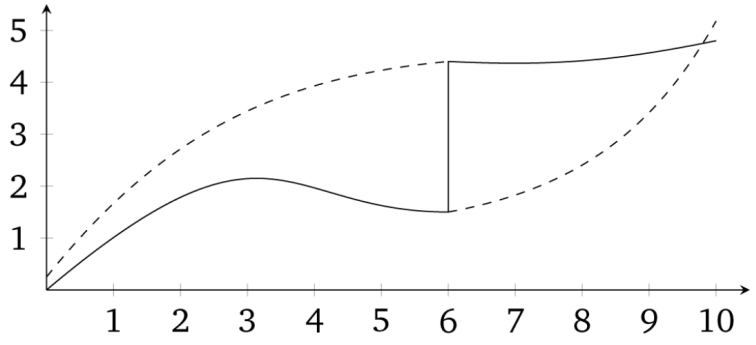
$$\lim_{X_i \to c_0} \mathbf{P}(D_i = 1 | X_i = c_0) \neq \lim_{c_0 \leftarrow X_i} \mathbf{P}(D_i = 1 | X_i = c_0)$$

Fuzzy RD: Continuity assumption

Observed data in Fuzzy RDD



Continuity assumption in Fuzzy RDD



Imbens and Lemieux (2008)

- Assume the conditional expectation of potential outcomes is smooth throughout the cutoff c_0
- What changes at c_0 is the treatment assignment probability

Fuzzy RDD: Estimation

Can estimate effects using a type of Wald estimator:

$$\delta_{\text{Fuzzy RDD}} = \frac{\lim_{x \downarrow c_0} \mathbf{E}[Y|X=x] - \lim_{x \uparrow c_0} \mathbf{E}[Y|X=x]}{\lim_{x \downarrow c_0} \mathbf{E}[D|X=x] - \lim_{x \uparrow c_0} \mathbf{E}[D|X=x]}$$

Fuzzy designs are numerically equivalent and conceptually similar to IV:

- "Reduced form" numerator: "jump" in regression of the outcome on the running variable X

Similar IV assumptions that we will discuss in the next section

- Monotonicity: X crossing a threshold cannot simultaneously cause some units to take up and others to reject D
- Excludability: X crossing the cutoff cannot impact Y except through impacting the receipt of D

Under these assumptions:

$$\delta_{\text{Fuzzy RDD}} = \mathbf{E}[Y_i(1) - Y_i(0)|T = c, X = c]$$

$T = c$ are units that receive treatment when $X > c$, but not otherwise.

- This is a **local ATE (LATE)**; an effect for a subpopulation locally at $X = c_0$

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Challenges

Challenges to RDD

Continuity assumption required for RDD. This assumption may not hold if:

- Manipulation on the running variable
- Endogeneity of the cutoff

Most robustness is aimed at building credibility around these

Manipulating running variable

Treatment is not as good as randomly assigned around the cutoff, c_0, when agents are able to manipulate their running variable scores. This happens when:

1. The assignment rule is known in advance
2. Agents are interested in adjusting
3. Agents have time to adjust
4. Administrative quirks like nonrandom heaping along the running

$\Longrightarrow$ Looking for evidence that people are *choosing* their X_i values, so as to barely receive/avoid treatment

- If people who sort along their running variable have different potential outcomes, your treatment effect estimation may be biased
- Testable implication: if there's no manipulation, then the density of X should be smooth around the cutoff

McCrary Density Test

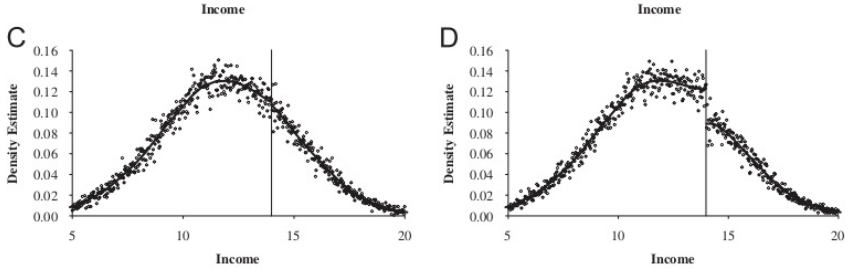
Null hypothesis: density is continuous at the cutoff point

- Alternative hypothesis: the density increases at the cutoff (may happen if the treatment is desirable)

Method:

- Count observations for a chosen bin (needs multiple units in other words per bin)
- Estimate nonlinear OLS model with quadratics in the running variable on the counts
- If unable to reject the null, that's a sign of manipulation

Simulated data:



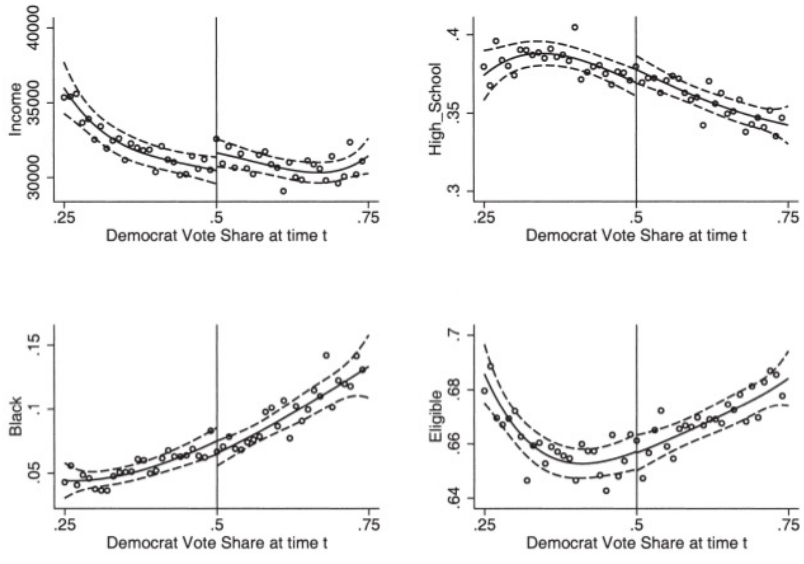
Placebo tests

For continuity assumption to hold, the only thing that causes the outcome to change abruptly at the cutoff is the treatment.

$\Longrightarrow$ There should not be discontinuous changes in average values of reasonably chosen covariates around the cutoff

Placebo test:

- If your continuity assumption is reasonable, Make sure there is no effect around reasonable chosen pre-treatment variables



Heaping

What is the causal effect of heightened medical care for premature babies on infant mortality?

- Babies with very low birth weights ($<$ 1500g) are automatically admitted to the NICU

Problem with using this cutoff: heaping

- Unlikely babies are naturally born at intervals of 100g
- Likely that there is some manipulation of running variable X

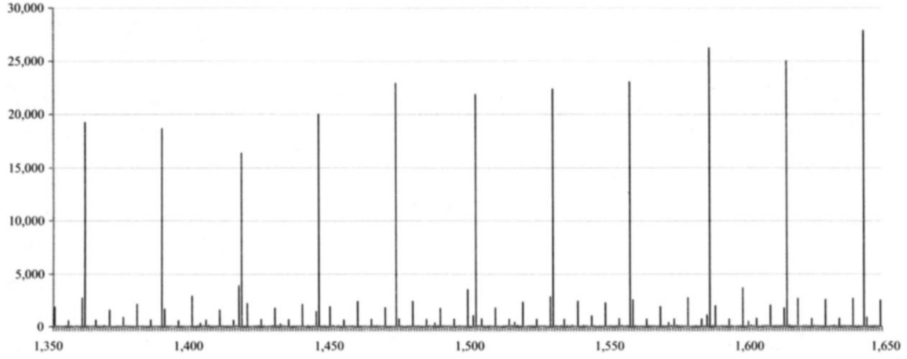


FIGURE I
Frequency of Births by Gram: Population of U.S. Births between 1,350 and 1,650 g
NCHS birth cohort linked birth/infant death files, 1983–1991 and 1995–2003, as described in the text.

Donut RDD

Density tests cannot detect heaping, because of low power around the cut-off

Solution: Drop units in the vicinity of the cutoff and re-estimate the model (called "donut hole")

- Because of dropped units, the estimated parameter is an even more unusual local average treatment effect
- But this method allows for the possibility that units at the heap differ markedly due to selection bias